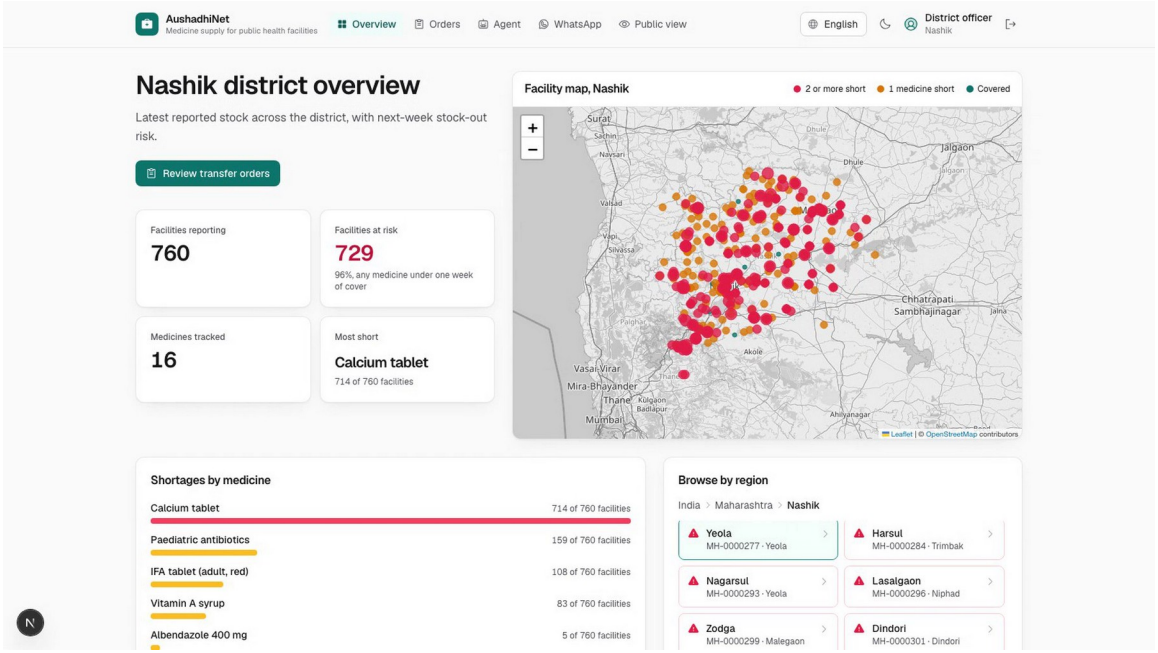
Build with AI: Code for Communities · Problem Statement 03

AushadhiNet

Early warning and rebalancing for public-health medicine supply, reported over WhatsApp.



Live on Google Cloud Run: aushadhinet-frontend-449585498230.asia-south1.run.app

Code: github.com/akshat333-debug/AushadhiNet

The problem

Clinics run dry while neighbours hold spare stock

Primary health centres run out of ORS, zinc, iron tablets and antibiotics.

Stock is reported late, on paper, and rarely compared across facilities.

Shortage and surplus sit a few kilometres apart; nobody sees both in time.

Frontline nurses won't adopt yet another app.

How it works

Report → Forecast → Rebalance → Human approves

1 Report

Nurse sends stock on WhatsApp in 7 languages; unsure reads are confirmed first.

2 Forecast

LightGBM predicts next-week demand and stock-out probability per medicine.

3 Rebalance

Google OR-Tools proposes transfers from nearby surplus to at-risk facilities.

4 Approve

Officer approves; both facilities notified. The AI agent can explain and draft, never approve.

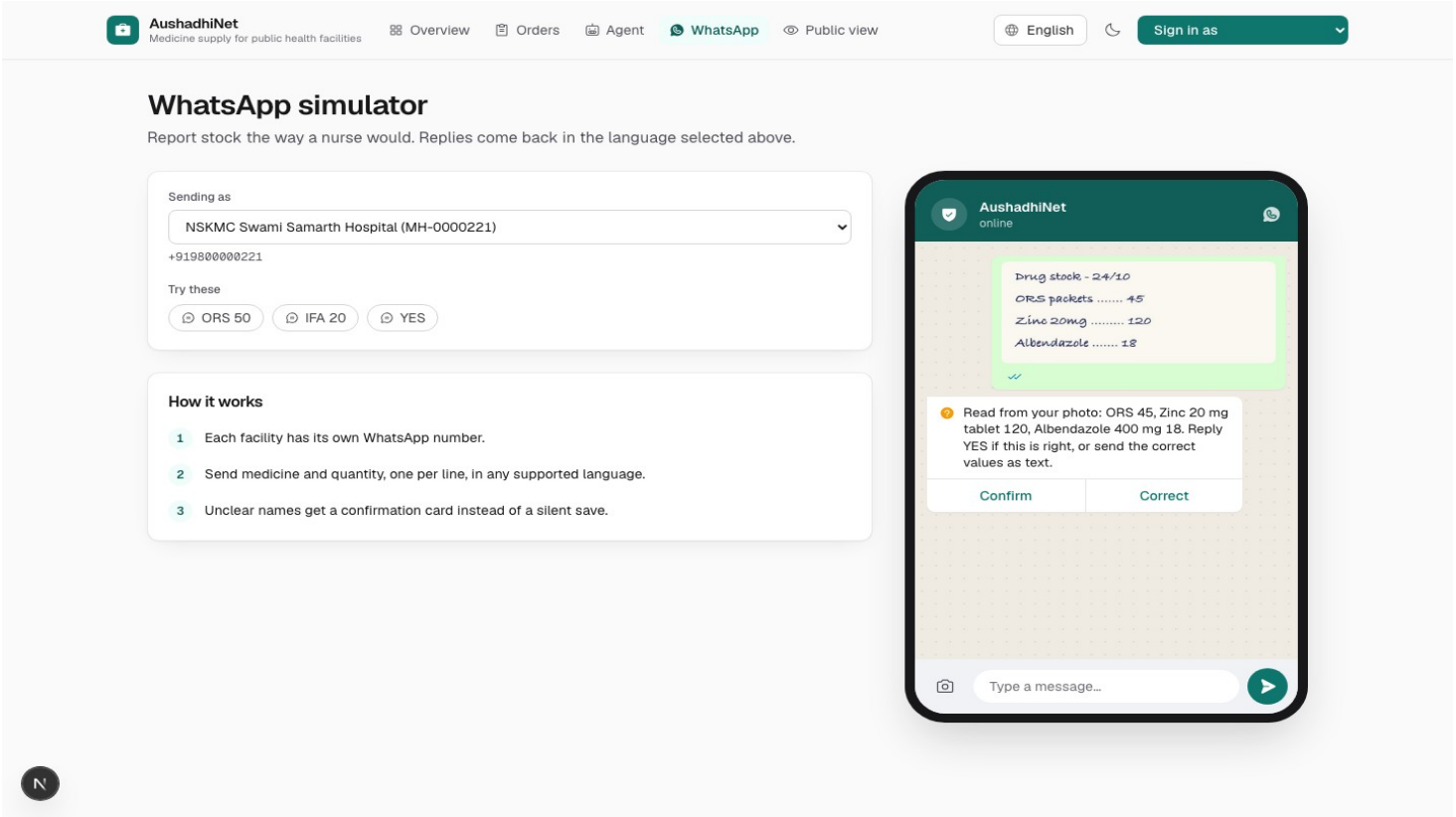
Nurse channel

WhatsApp, no new app

Text in any script, or a photo of the register

Gemini reads every line of a photo

Nothing is saved until the nurse confirms



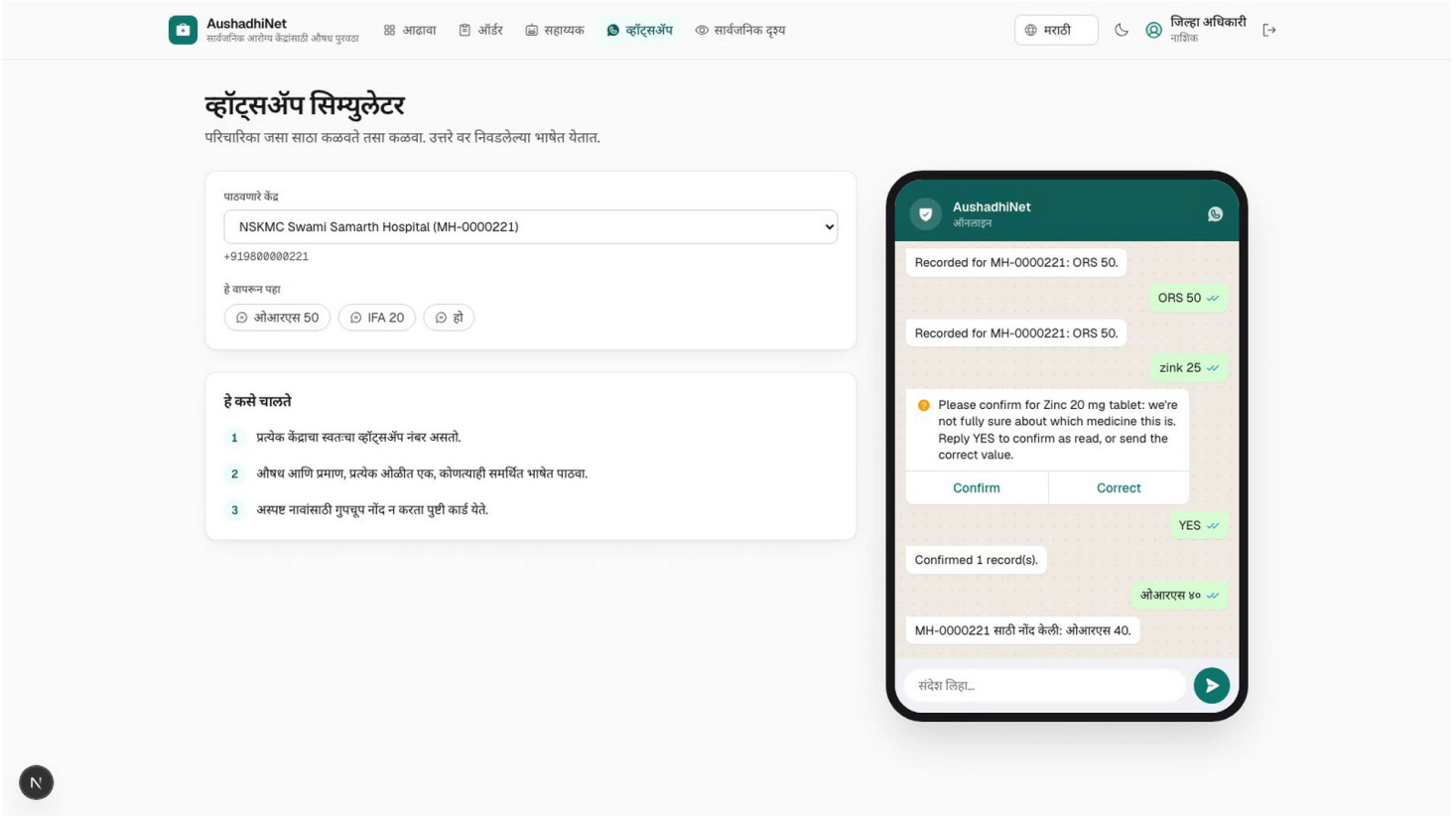
Multilingual

Seven Indian languages

English, Hindi, Marathi, Bengali, Tamil, Telugu, Kannada

UI, medicine names, replies and agent answers

Translations need native-speaker review



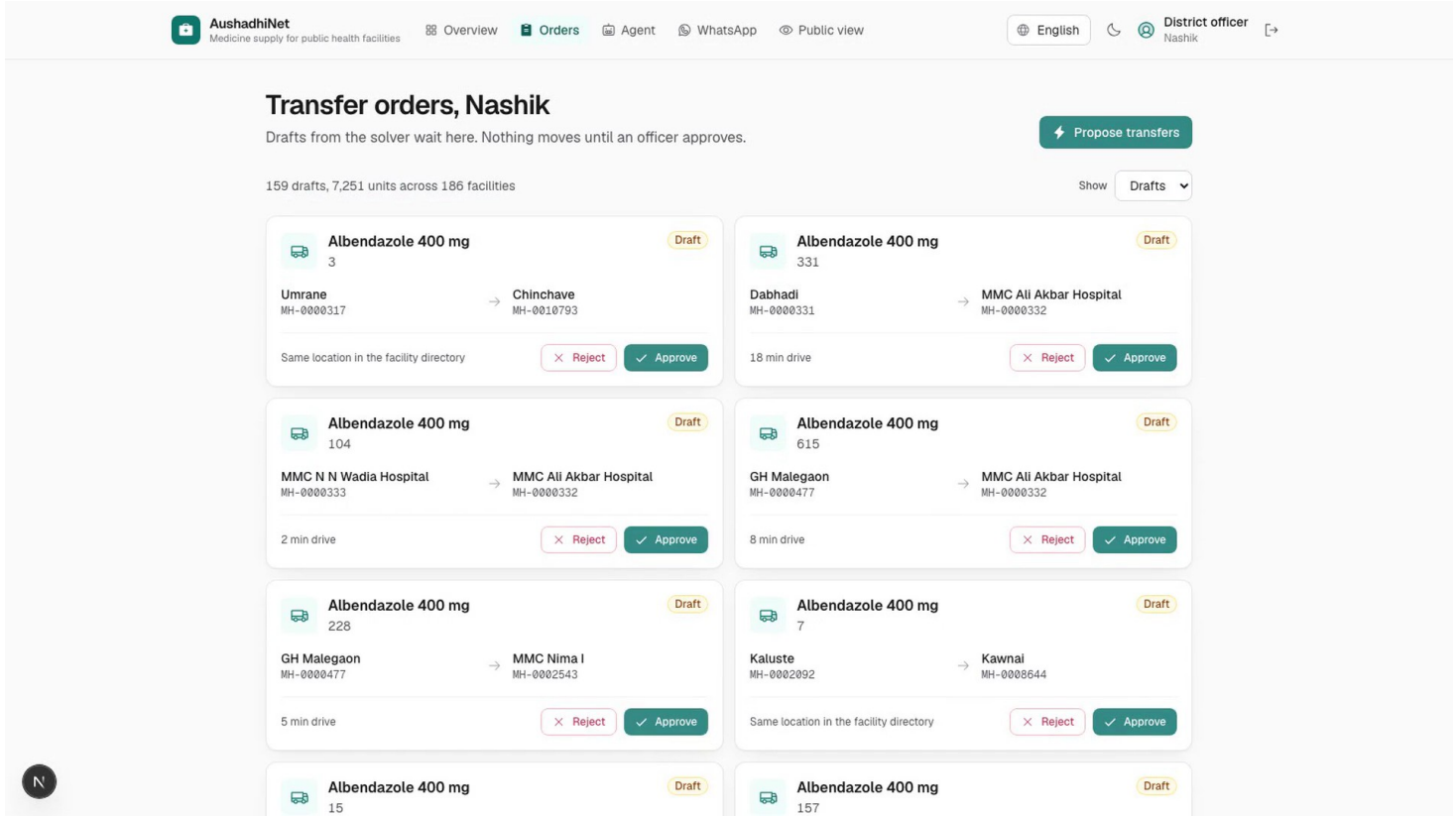
Officer view

Transfer orders, human in the loop

Solver drafts transfers (OR-Tools)

Officer approves or rejects each

Orders signed with Cloud KMS; both ends notified



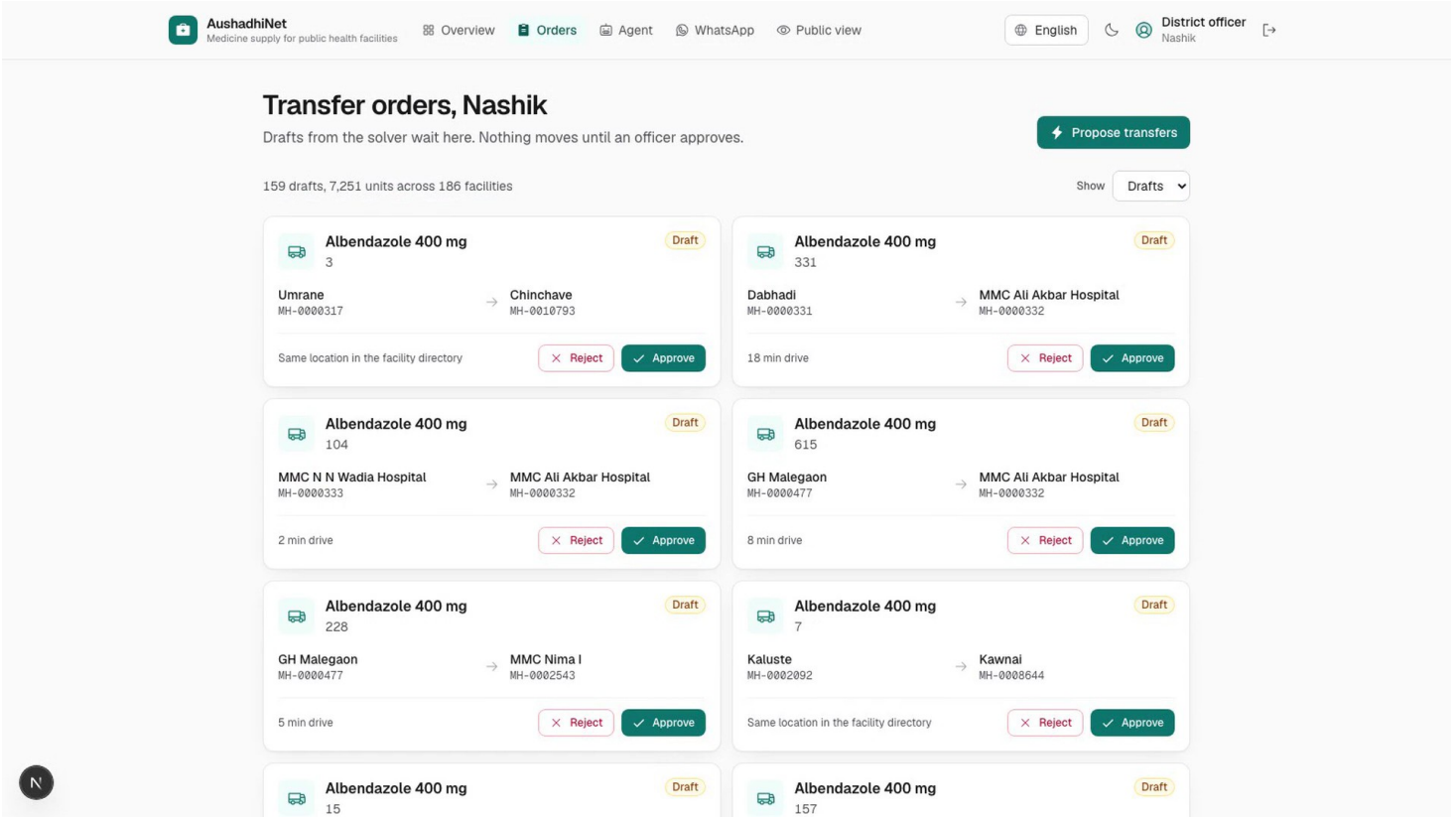
Guardrailed agent

Explains, never acts

Answers from real forecast and stock data

Jurisdiction-scoped per officer

Refuses prompt-injection to approve orders



Results

Measured on a frozen, held-out protocol

0.871

Forecast WAPE on real HMIS data
vs 1.078 naive (19% better)

−6.0%

Stock-out weeks in 26-week replay
(synthetic Nashik ledger)

760

Nashik facilities × 16 HMIS medicines in the pilot

7

Languages end to end

Test window frozen before tuning; first model (L2) lost to naive and stays in the run log.

Impact

AushadhiNet by the numbers

760

facilities monitored in Nashik pilot

16

essential HMIS medicines tracked

729

facilities flagged at risk before stock-out

159

transfer orders drafted by OR-Tools

7,251

units rebalanced across 186 facilities

19%

lower forecast error than naive (WAPE 0.871)

−6%

stock-out weeks in 26-week replay

7

Indian languages, end to end

349

automated tests (335 backend + 14 E2E, passing on Cloud Run)

Google AI in the loop

Gemini reads register photos

100%

quantity read correctly

98.1%

medicine matched

4

scripts: English, Devanagari, Tamil, Bengali

Measured on 206 rendered register lines (synthetic text, not real handwriting photos).

Every photo read is confirmed by the nurse before it is saved: a model can be confidently wrong.

Next: measure on real register photos in the pilot.

Rollout

From one block to a state, on Google Cloud

Pilot

1 Nashik block: nurses' WhatsApp numbers, real register photos, 1 officer approving transfers

District

760 facilities: forecasts and transfer drafts weekly; measure stock-out weeks against last year

State

One GCP project per state (Terraform), so raw facility data stays in the state; federated model updates only

Running cost today: Cloud Run scales to zero; the demo runs on the GCP free-trial credit with a budget alert.

Firebase sign-in and Twilio WhatsApp are wired in code and are the pilot's two configuration steps.

Tech

Running on Google Cloud

Cloud Run · Firestore · BigQuery · Pub/Sub · Cloud KMS · Secret Manager (Terraform)

Gemini (register photos) · Vertex AI embeddings · Google OR-Tools

LightGBM · FastAPI · Next.js 15 · 349 automated tests

Thank you

Earlier warnings. Smarter sharing. People in control.

Live on Google Cloud Run: aushadhinet-frontend-449585498230.asia-south1.run.app

Next: pilot in one Nashik block with real registers.